

Image Segmentation

Computer Vision Tasks

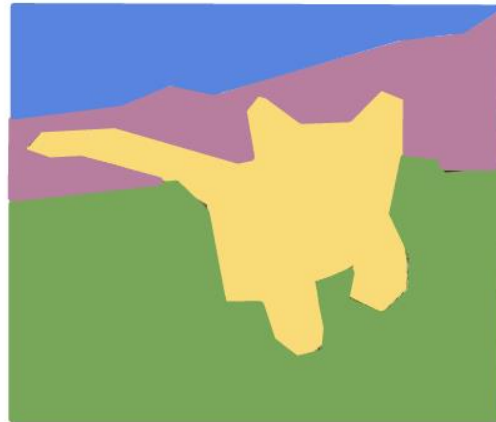
Classification



CAT

No spatial extent

Semantic Segmentation



**GRASS, CAT,
TREE, SKY**

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation

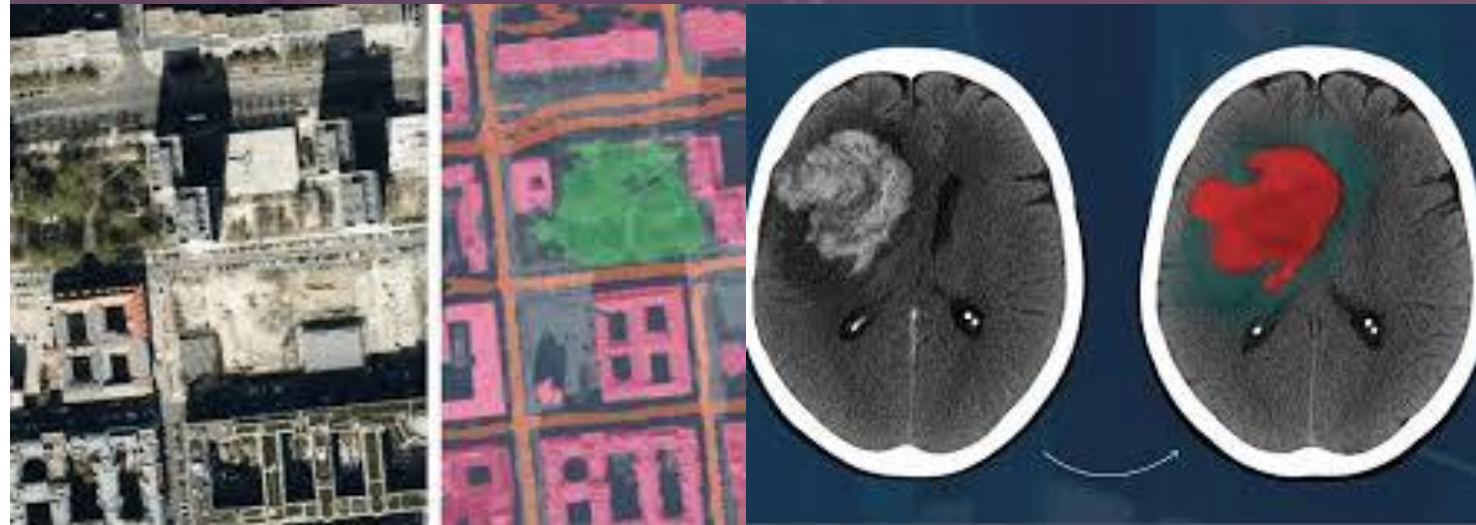


DOG, DOG, CAT

[This image](#) is [CC0 public domain](#)

Image Segmentation

- Process of dividing an image into meaningful regions
- Assigns a **label to every pixel**
- **Types:**
 - Semantic segmentation
 - Instance segmentation
 - Panoptic segmentation
- **Applications:**
 - Medical imaging
 - Autonomous driving
 - Satellite imagery
- **Challenges:**
 - Complex visual patterns
 - Limited labeled data
 - Class imbalance
 - Boundary precision issues
 - High computational cost



Semantic Segmentation

Classification



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No spatial extent

Semantic Segmentation



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TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

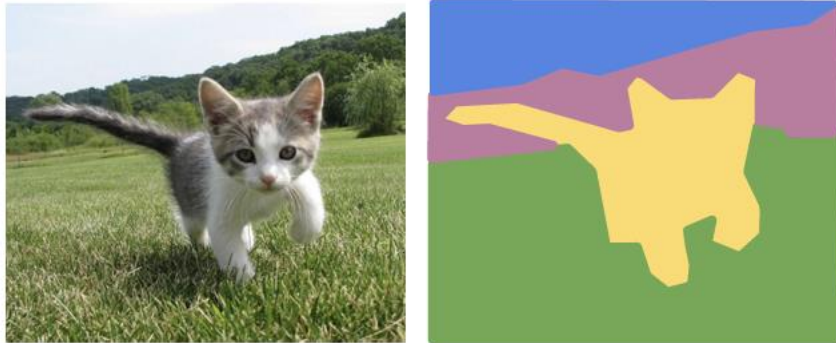
Multiple Object

Instance Segmentation



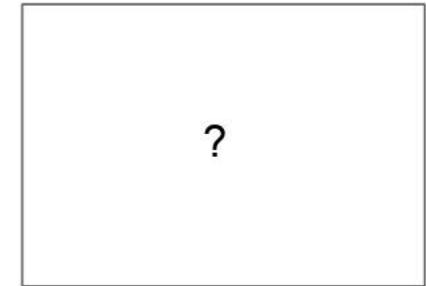
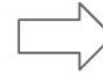
DOG, DOG, CAT

Semantic Segmentation: The Problem



GRASS, CAT,
TREE, SKY, ...

Paired training data: for each training image, each pixel is labeled with a semantic category.



At test time, classify each pixel of a new image.

Semantic Segmentation Idea: Sliding Window

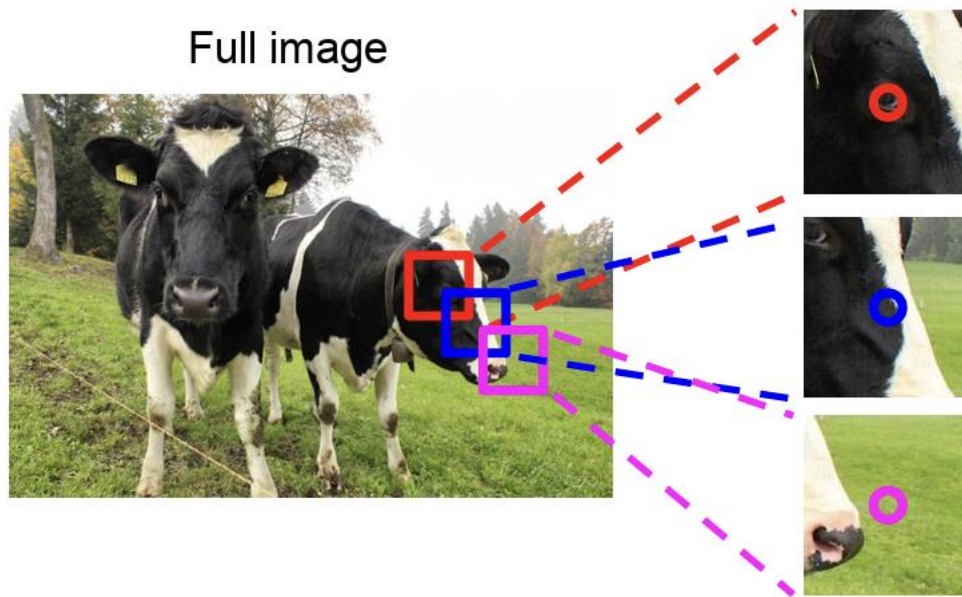
Full image



Impossible to classify without context

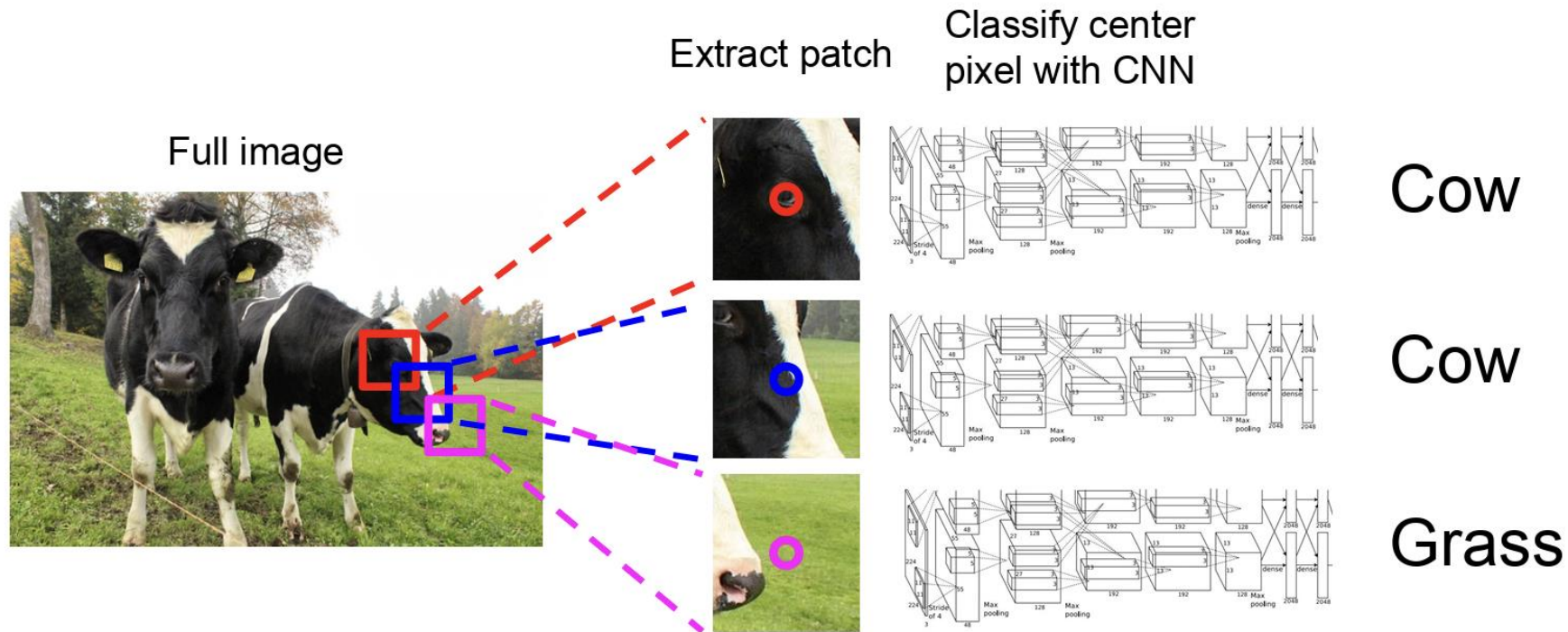
Q: How do we include context?

Semantic Segmentation Idea: Sliding Window



Q: how do we model this?

Semantic Segmentation Idea: Sliding Window

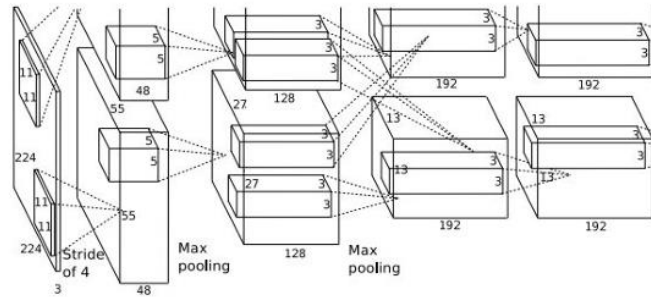


Problem: Very inefficient! Not reusing shared features between overlapping patches

Farabet et al, "Learning Hierarchical Features for Scene Labeling," TPAMI 2013
Pinheiro and Collobert, "Recurrent Convolutional Neural Networks for Scene Labeling", ICML 2014

Semantic Segmentation Idea: Convolution

Full image

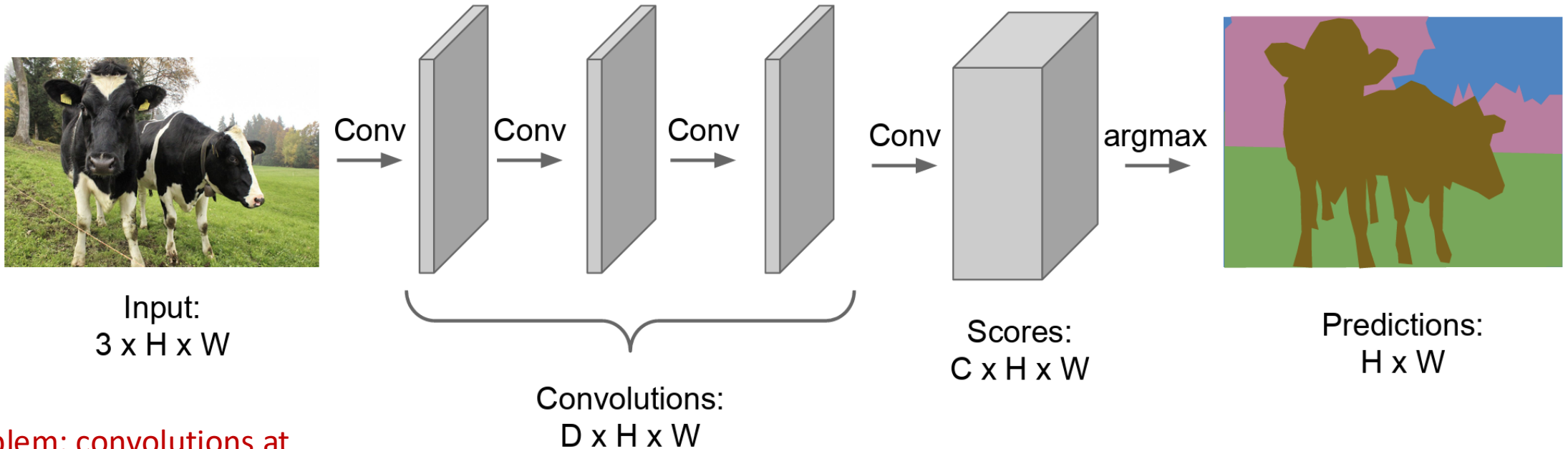


An intuitive idea: encode the entire image with conv net, and do semantic segmentation on top.

Problem: classification architectures often reduce feature spatial sizes to go deeper, but semantic segmentation requires the output size to be the same as input size.

Semantic Segmentation Idea: Fully Convolutional

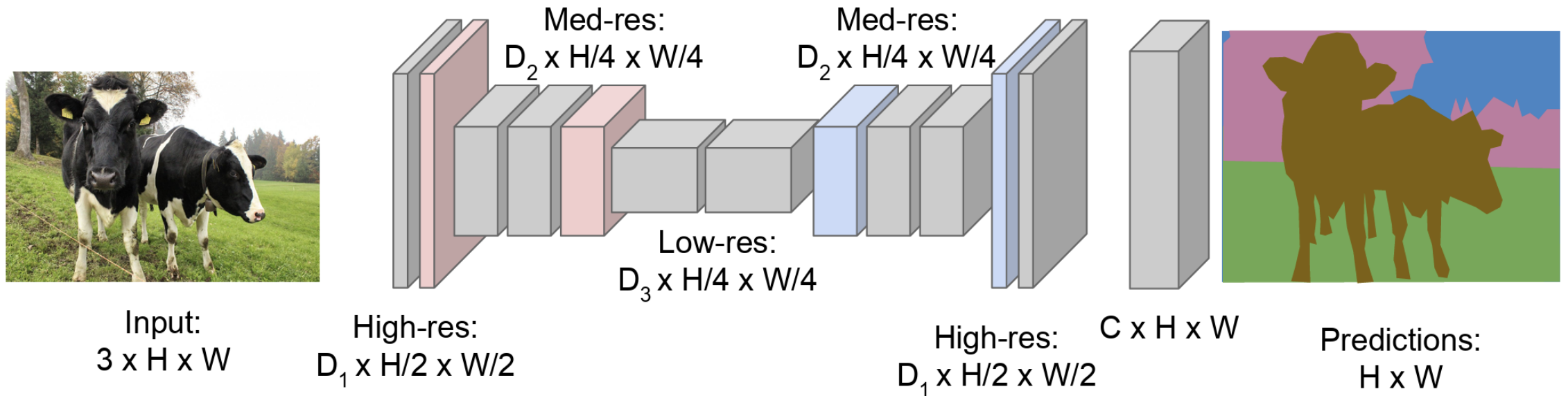
Design a network with only convolutional layers without downsampling operators to make predictions for pixels all at once!



Problem: convolutions at original image resolution will be very expensive

Semantic Segmentation Idea: Fully Convolutional

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!



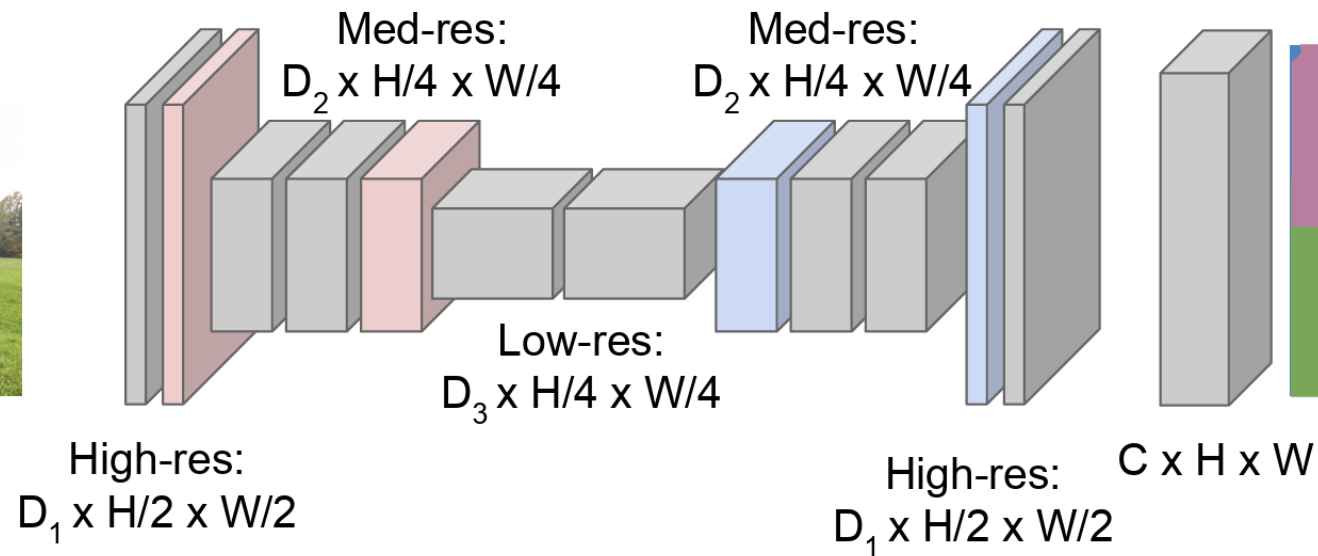
Semantic Segmentation Idea: Fully Convolutional

Downsampling:
Pooling, strided
convolution



Input:
 $3 \times H \times W$

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!



Upsampling:
???



Predictions:
 $H \times W$

In-Network upsampling: “Unpooling”

Nearest Neighbor

1	2
3	4



1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Input: 2 x 2

Output: 4 x 4

“Bed of Nails”

1	2
3	4

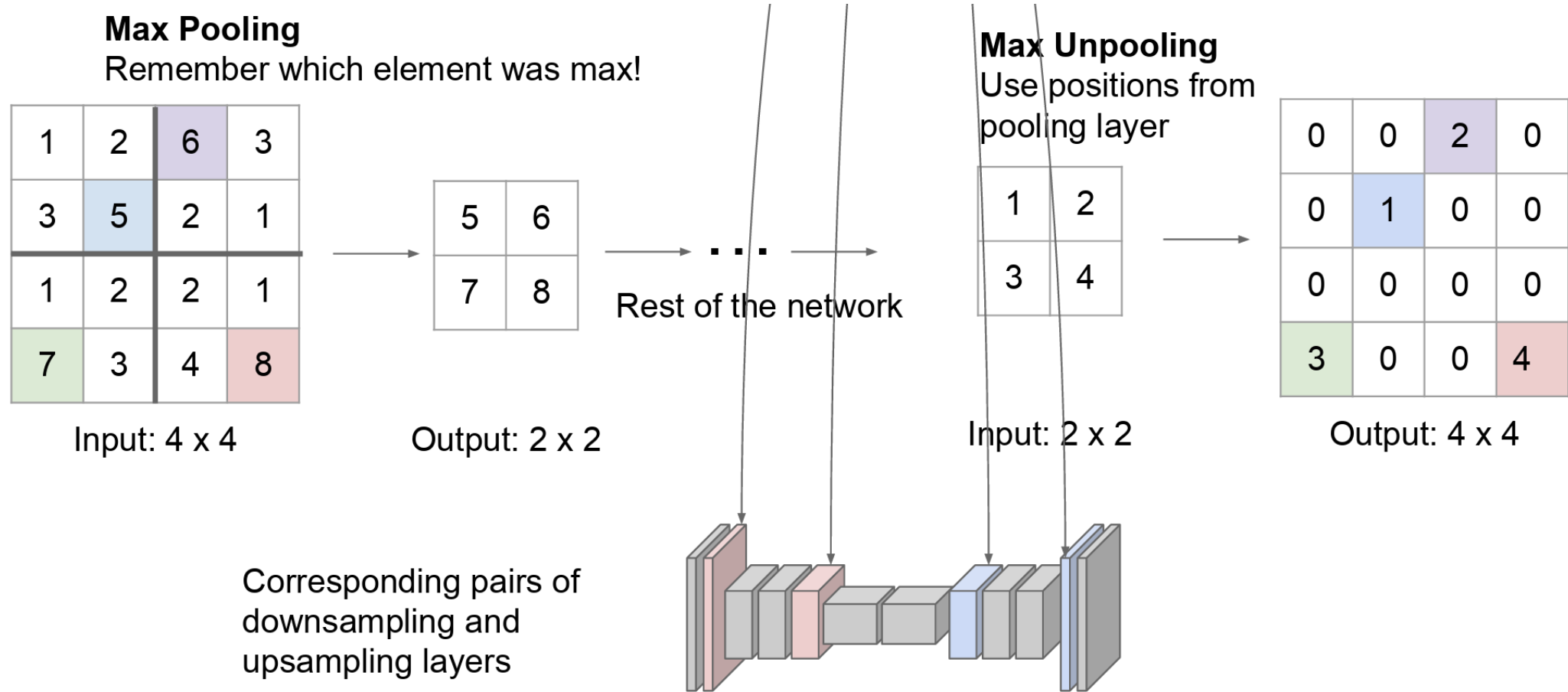


1	0	2	0
0	0	0	0
3	0	4	0
0	0	0	0

Input: 2 x 2

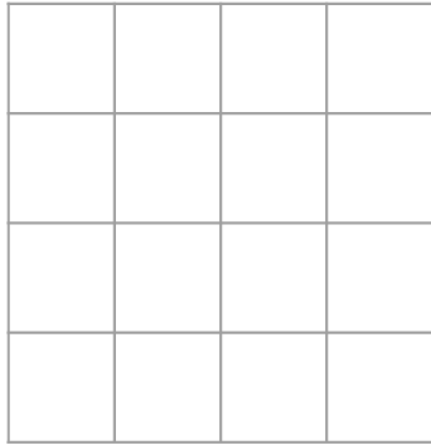
Output: 4 x 4

In-Network upsampling: “Max Unpooling”

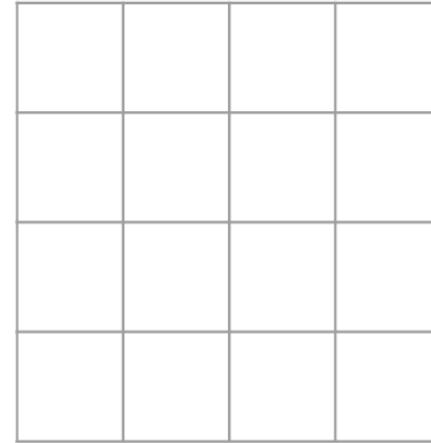


Learnable Upsampling

Recall: Normal 3 x 3 convolution, stride 1 pad 1



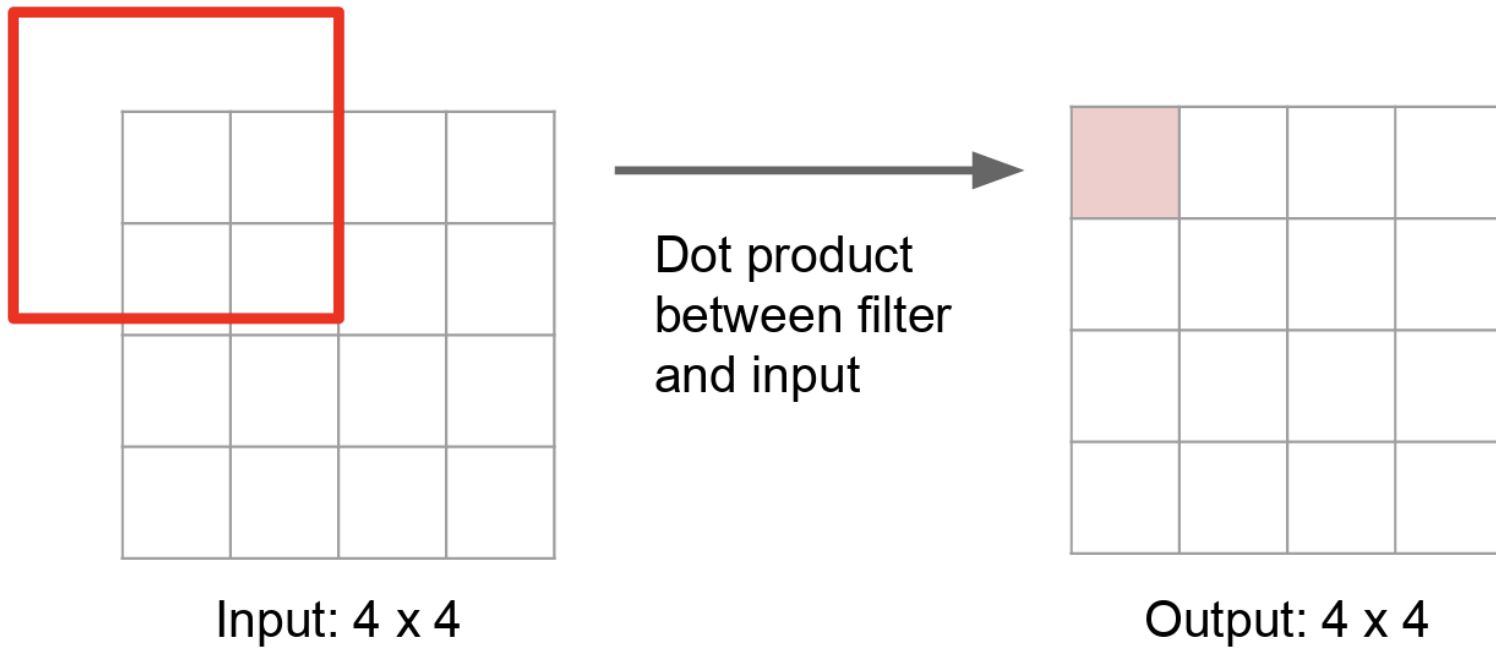
Input: 4 x 4



Output: 4 x 4

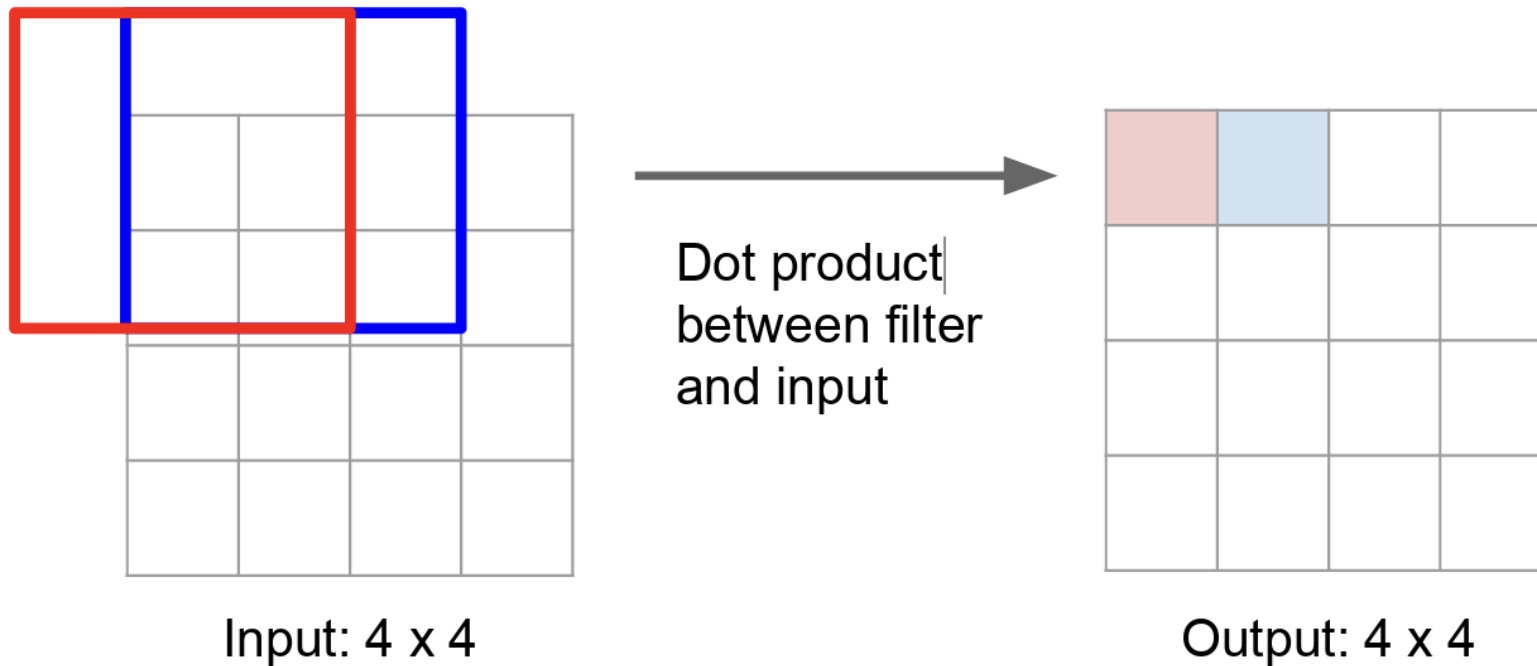
Learnable Upsampling

Recall: Normal 3 x 3 convolution, stride 1 pad 1



Learnable Upsampling

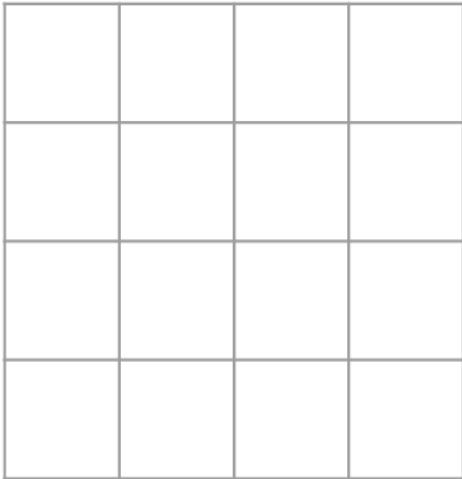
Recall: Normal 3 x 3 convolution, stride 1 pad 1



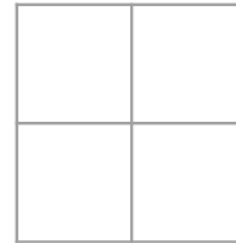
- Output matrix size = $\frac{n+2P-f}{s} + 1$

Learnable Upsampling

Recall: Normal 3 x 3 convolution, stride 2 pad 1



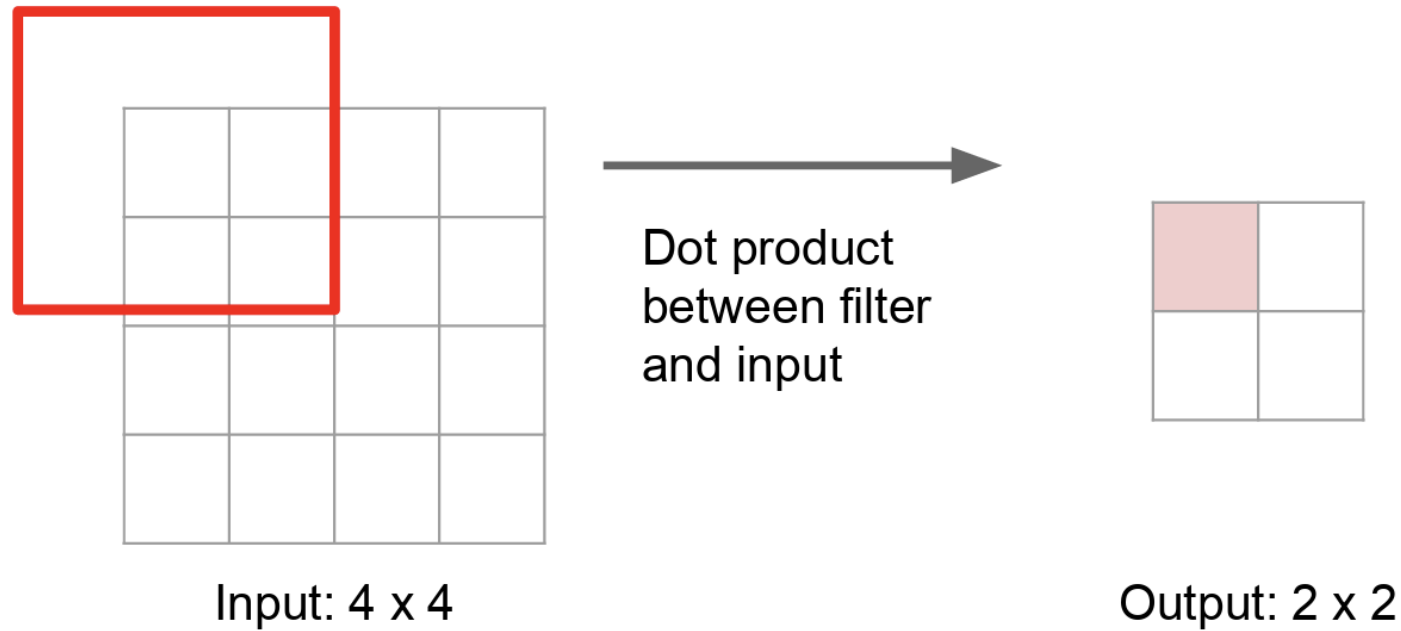
Input: 4 x 4



Output: 2 x 2

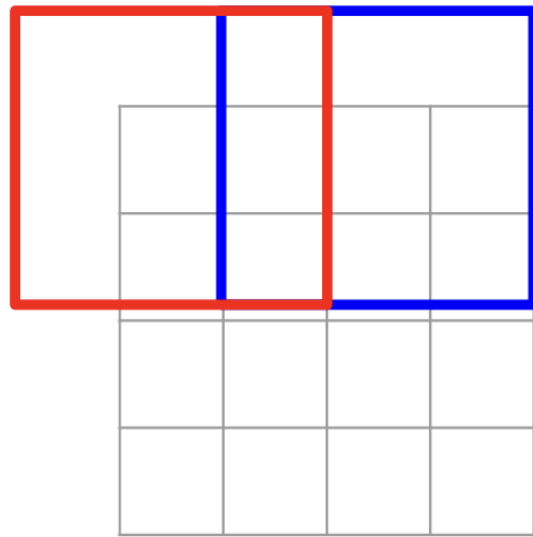
Learnable Upsampling

Recall: Normal 3 x 3 convolution, stride 2 pad 1



Learnable Upsampling

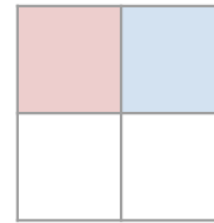
Recall: Normal 3 x 3 convolution, stride 2 pad 1



Input: 4 x 4



Dot product
between filter
and input



Output: 2 x 2

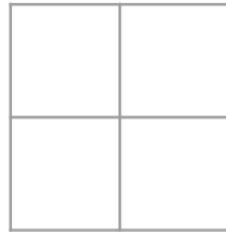
Filter moves 2 pixels in
the input for every one
pixel in the output

Stride gives ratio between
movement in input and
output

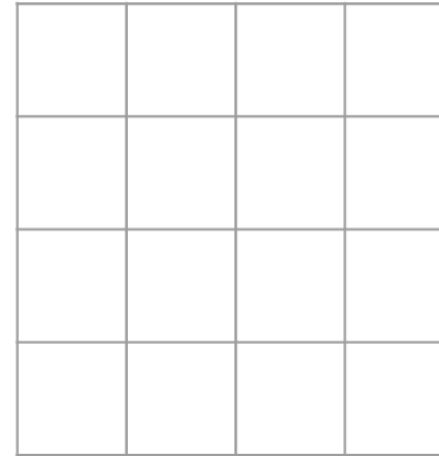
We can interpret strided
convolution as “learnable
downsampling”.

Learnable Upsampling: Transposed Convolution

3 x 3 **transposed** convolution, stride 2 pad 1



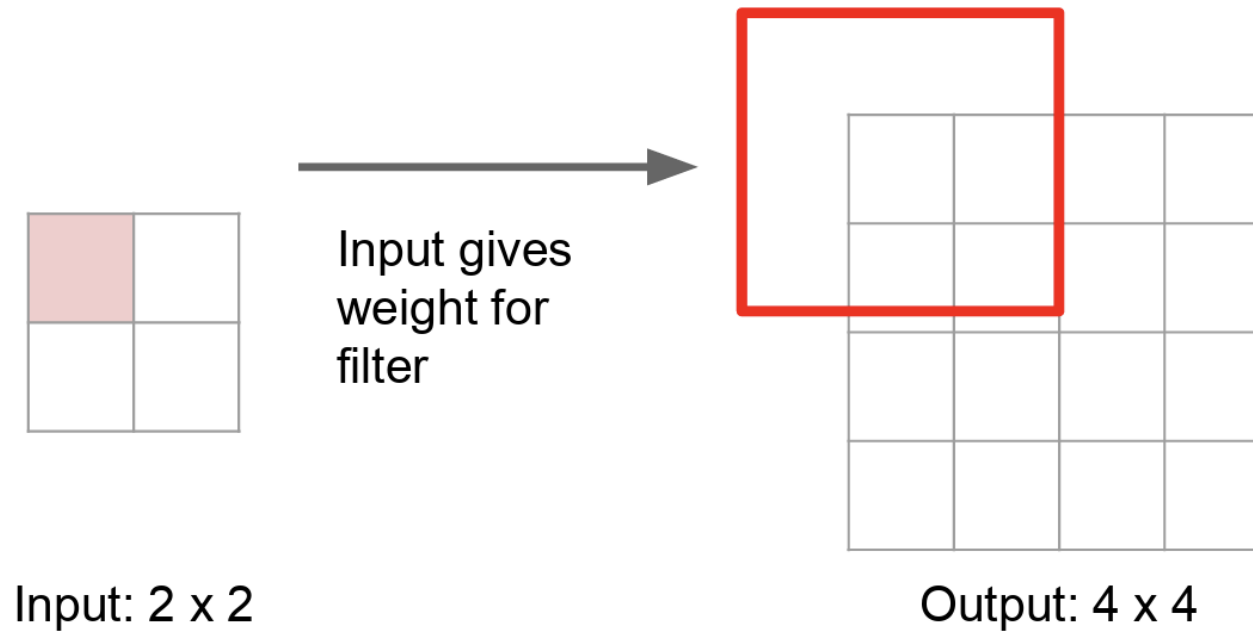
Input: 2 x 2



Output: 4 x 4

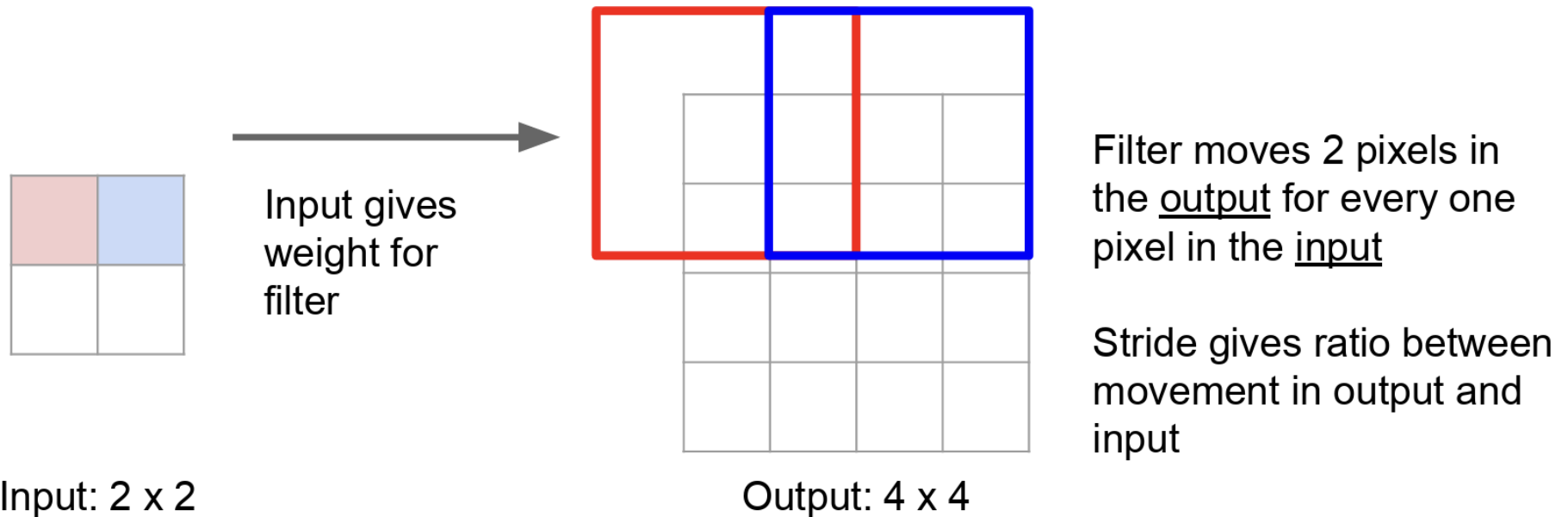
Learnable Upsampling: Transposed Convolution

3 x 3 **transposed** convolution, stride 2 pad 1

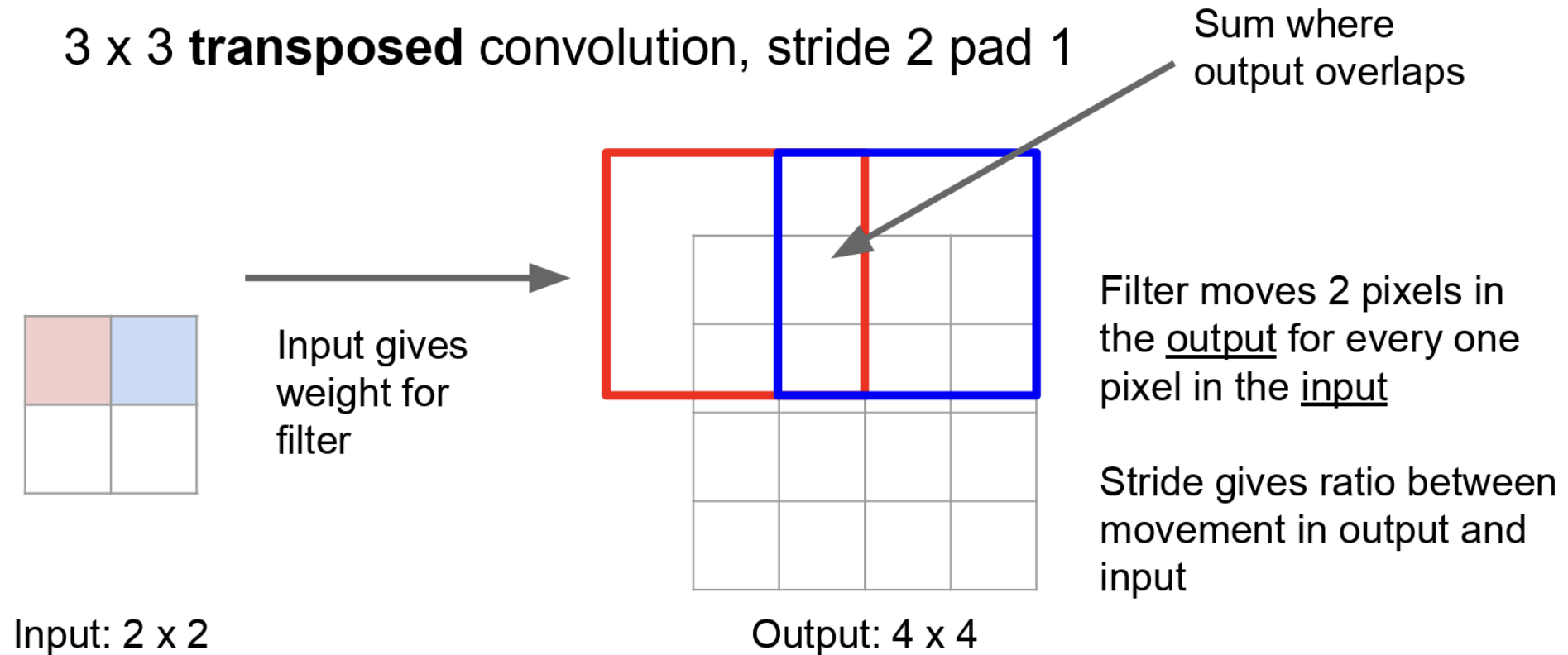


Learnable Upsampling: Transposed Convolution

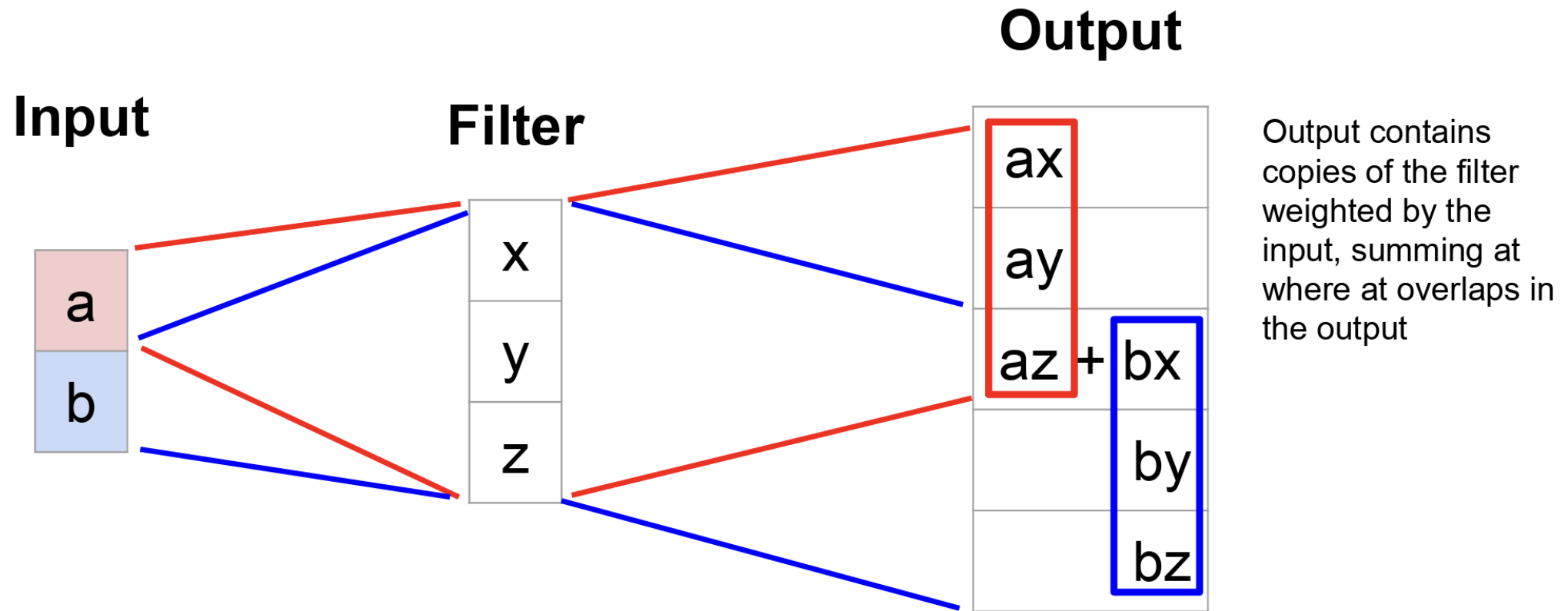
3 x 3 **transposed** convolution, stride 2 pad 1



Learnable Upsampling: Transposed Convolution



Learnable Upsampling: 1D Sample



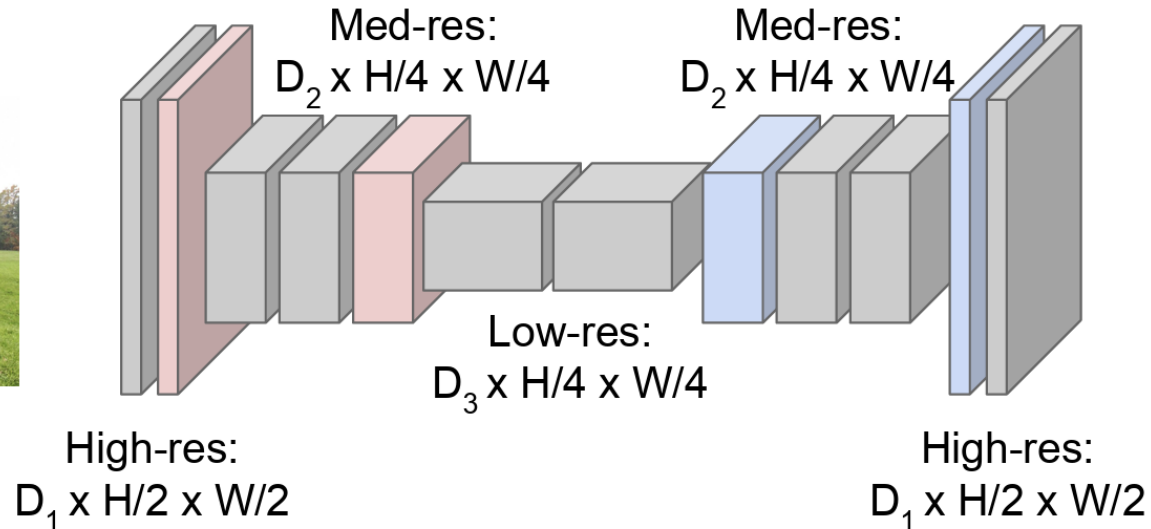
Semantic Segmentation Idea: Fully Convolutional

Downsampling:
Pooling, strided
convolution



Input:
 $3 \times H \times W$

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!



Upsampling:
Unpooling or strided
transposed convolution

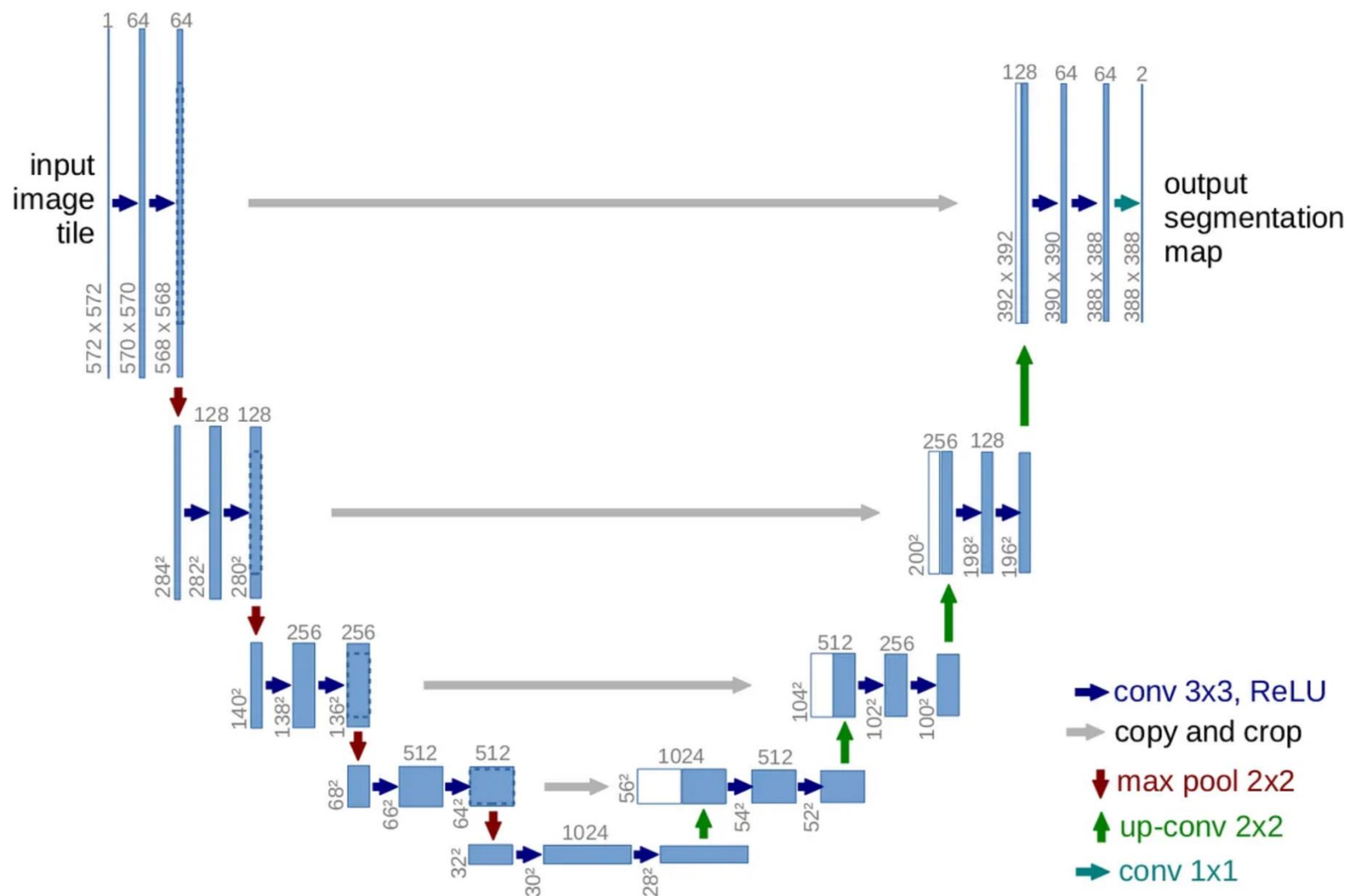


Predictions:
 $H \times W$

U-Net

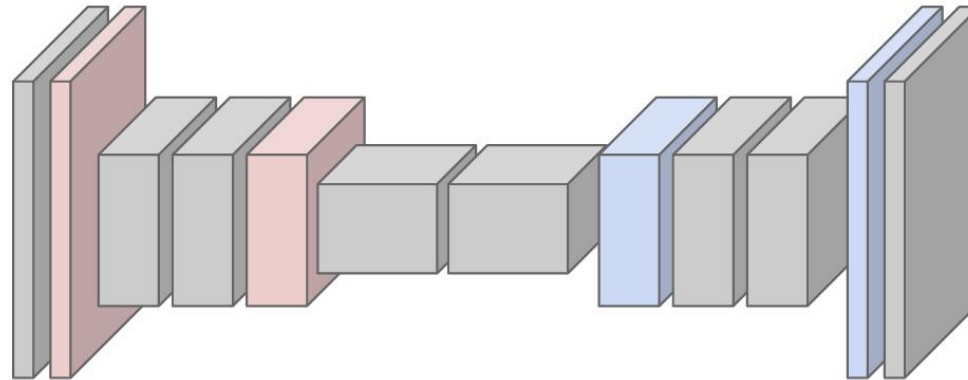
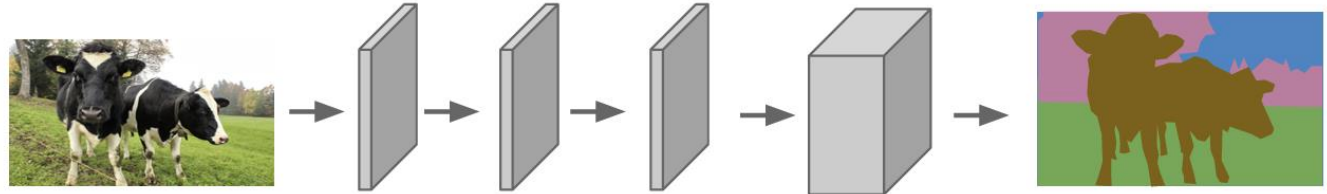
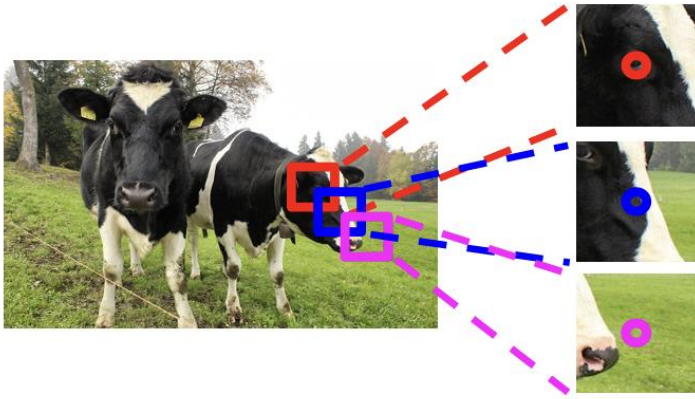


Input:
 $3 \times H \times W$



Predictions:
 $H \times W$

Semantic Segmentation: Summary

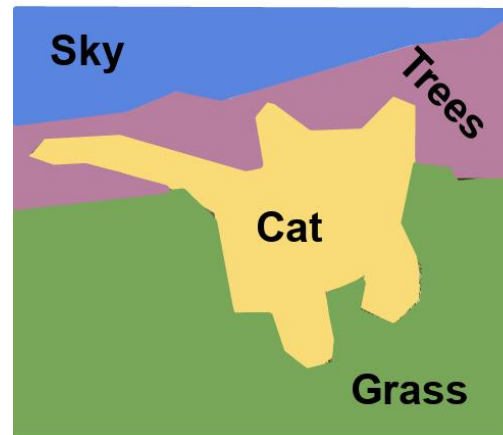


FCN vs U-Net

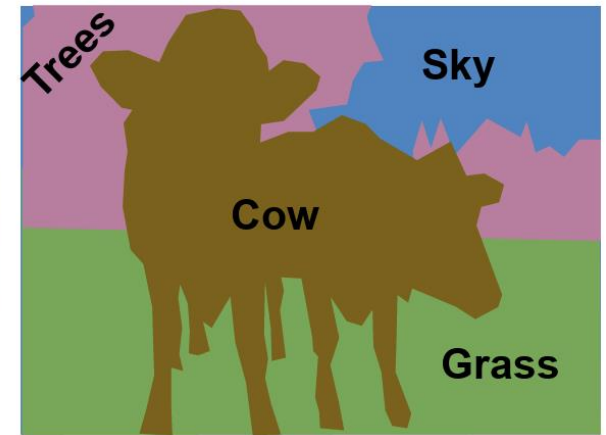
- **Fully Convolutional Networks (FCN)**
 - First deep learning model for semantic segmentation
 - Converts classification CNN → pixel-wise prediction
 - Uses **upsampling (deconvolution)**
 - Simple upsampling
- **U-Net**
 - Encoder–decoder architecture
 - Introduces **skip connections**
 - Designed for **biomedical segmentation**
 - feature fusion via skip connections

Semantic Segmentation

- Labels each pixel in the image with a category label
- Does not differentiate instances, only cares about pixels

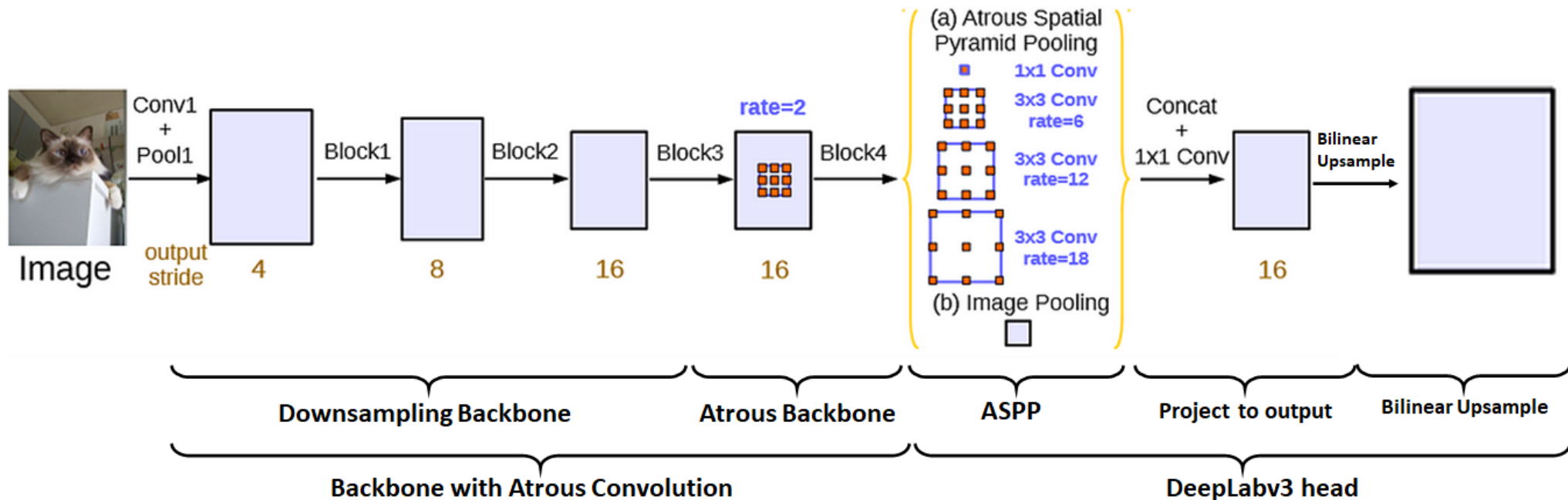


[This image is CC0 public domain](#)



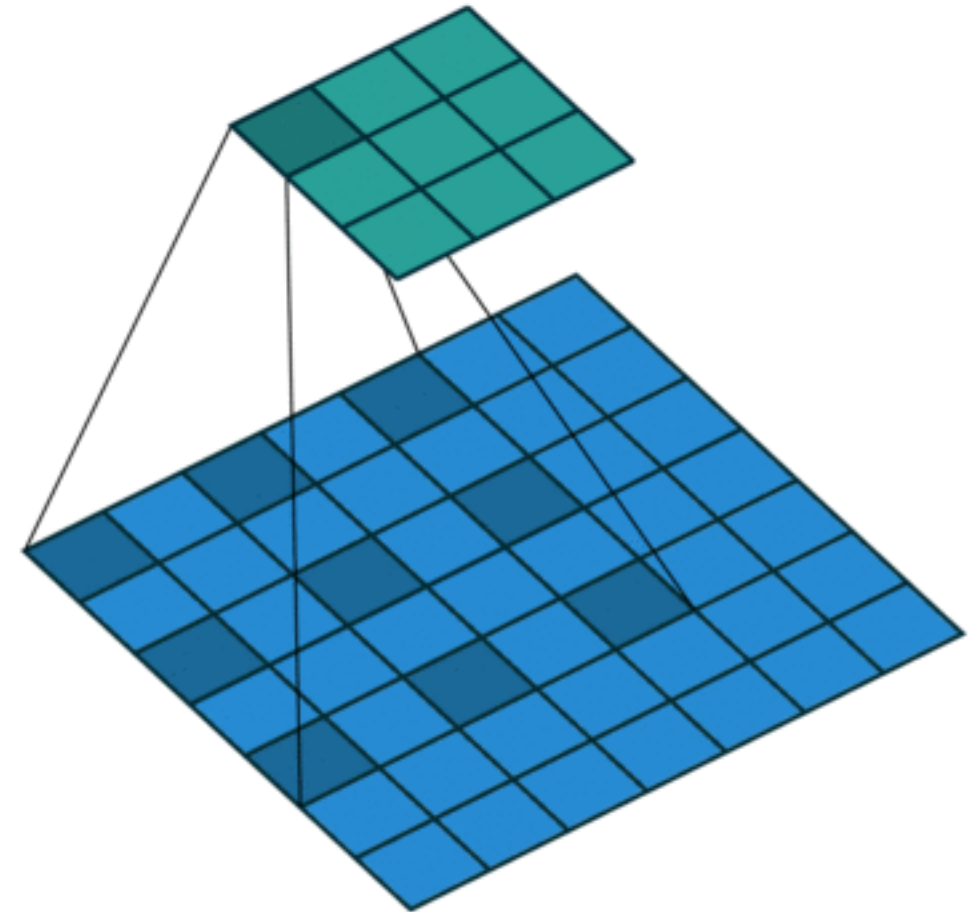
DeepLabv3+

- focuses on capturing **context at multiple scales** using a technique called **Atrous (Dilated) Convolution** and an enhanced module called **ASPP (Atrous Spatial Pyramid Pooling)** for **Semantic Segmentation**.
- **Backbone Network**
 - Typically uses pretrained CNNs like:
 - ResNet (ResNet-50 / ResNet-101)
 - Xception
 - Extracts feature maps from the input image.



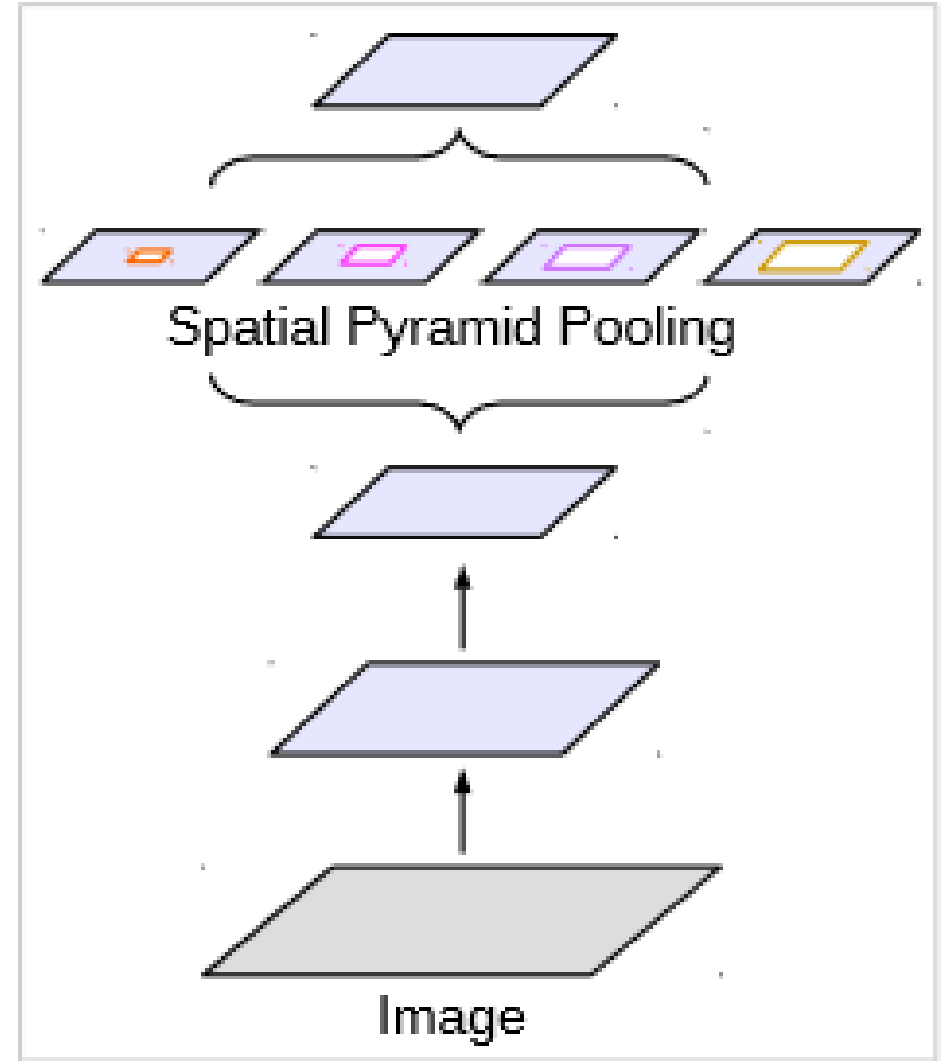
Atrous (Dilated) Convolution

- Expands the receptive field **without reducing resolution**. Helps capture fine details and global context simultaneously. Controlled by a **dilation rate**.
- **inserts zeros** into the convolution kernel to increase the size of the kernel without increasing the number of learnable parameters
- In the figure, we can see that the 3x3 atrous kernel has a 5x5 receptive field. If we were to stack up layers of atrous convolution, we would not only have a large receptive field, we would have more dense feature maps than with regular convolutions.



Atrous Spatial Pyramid Pooling (ASPP)

- Core innovation of DeepLabv3.
- Applies multiple parallel atrous convolutions with different dilation rates.
- Typical ASPP branches:
 - 1×1 convolution
 - 3×3 atrous conv (rate = 6)
 - 3×3 atrous conv (rate = 12)
 - 3×3 atrous conv (rate = 18)
 - Image-level pooling



Instance Segmentation

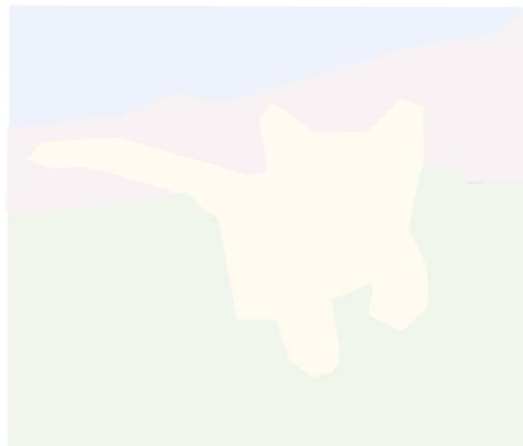
Classification



CAT

No spatial extent

Semantic Segmentation



GRASS, CAT,
TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

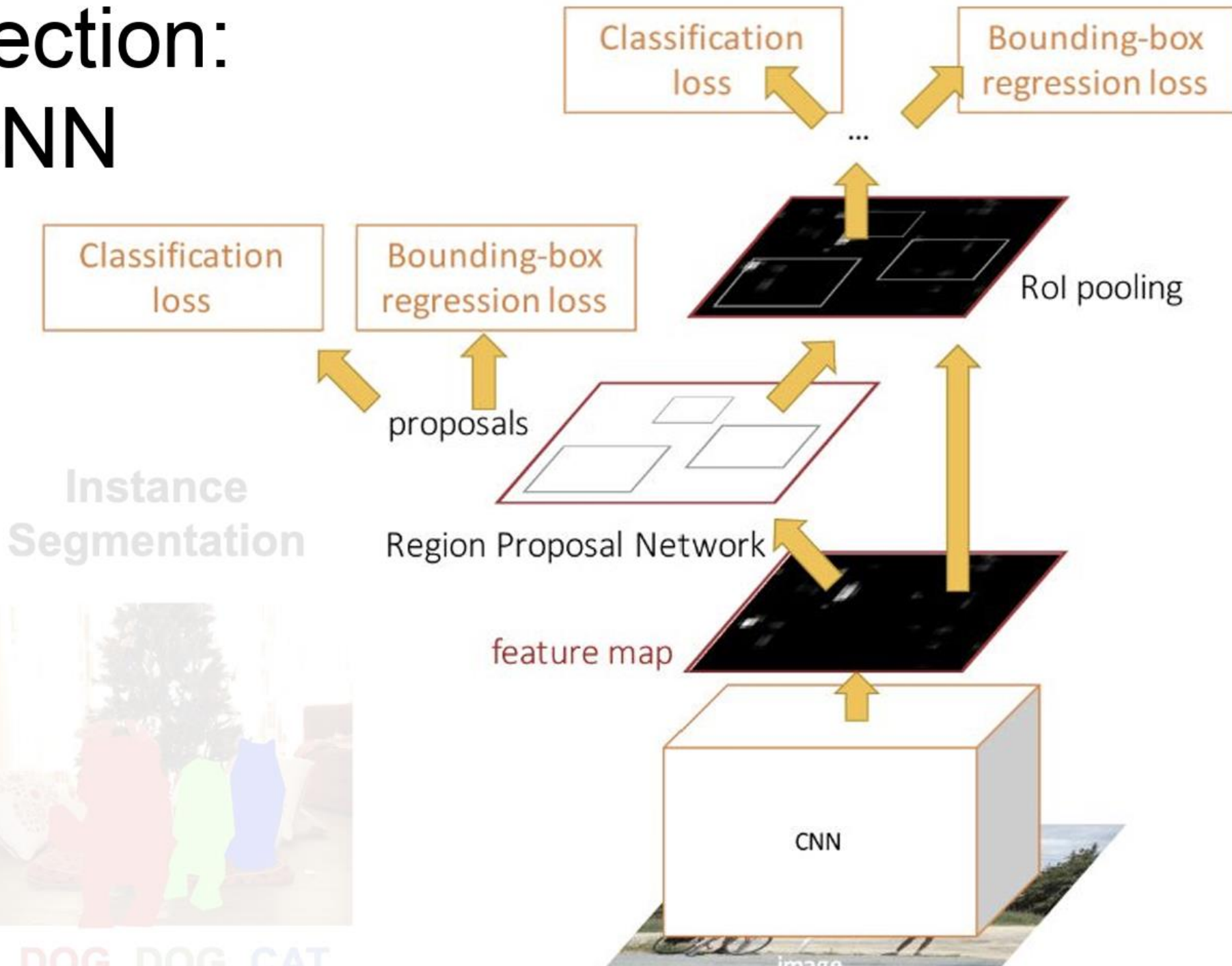
Multiple Object

Instance Segmentation

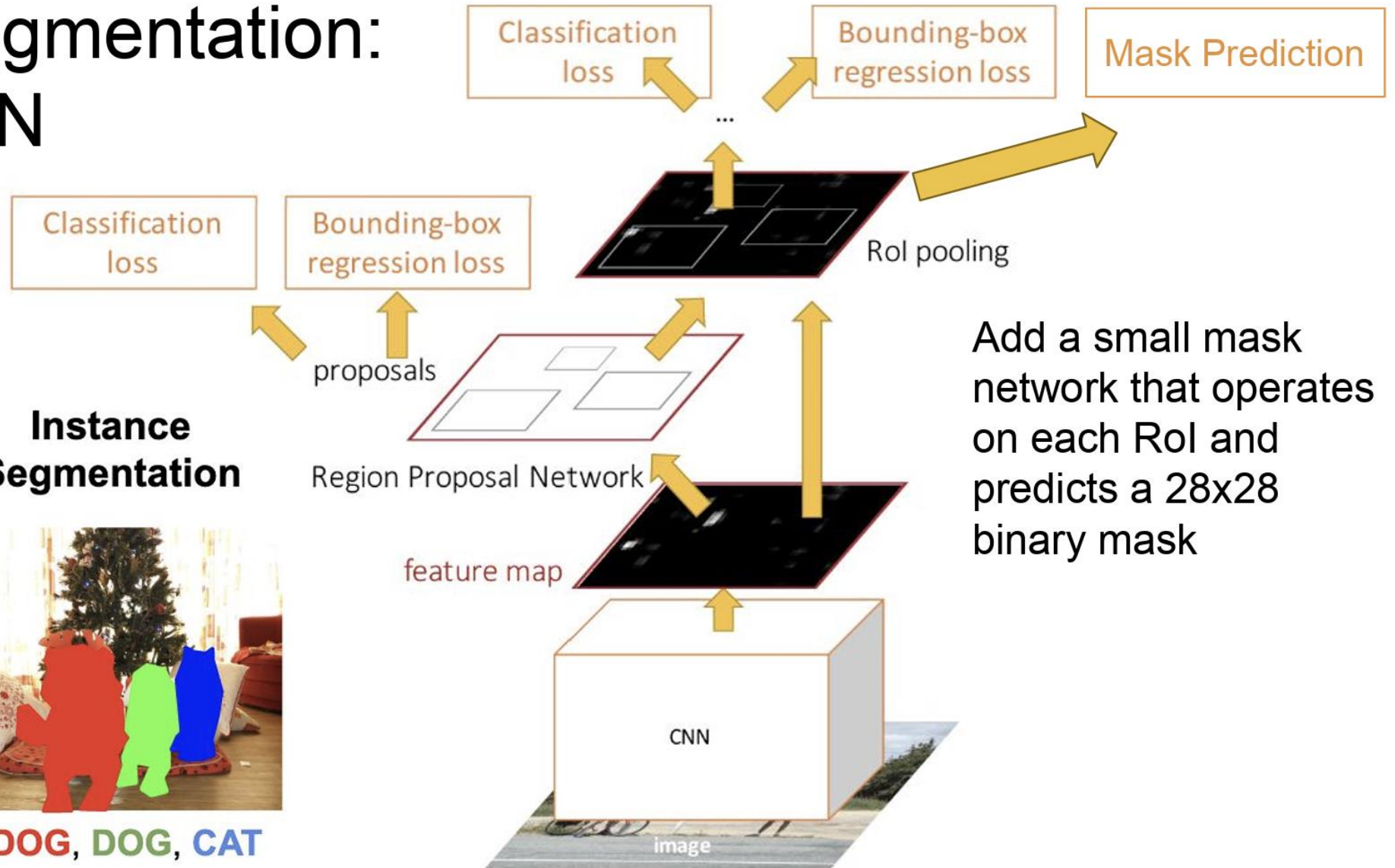
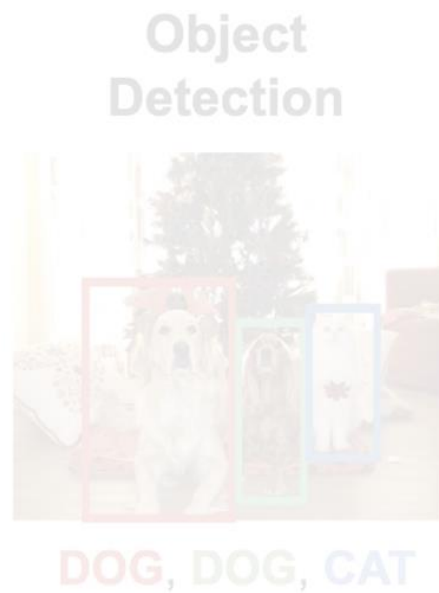


DOG, DOG, CAT

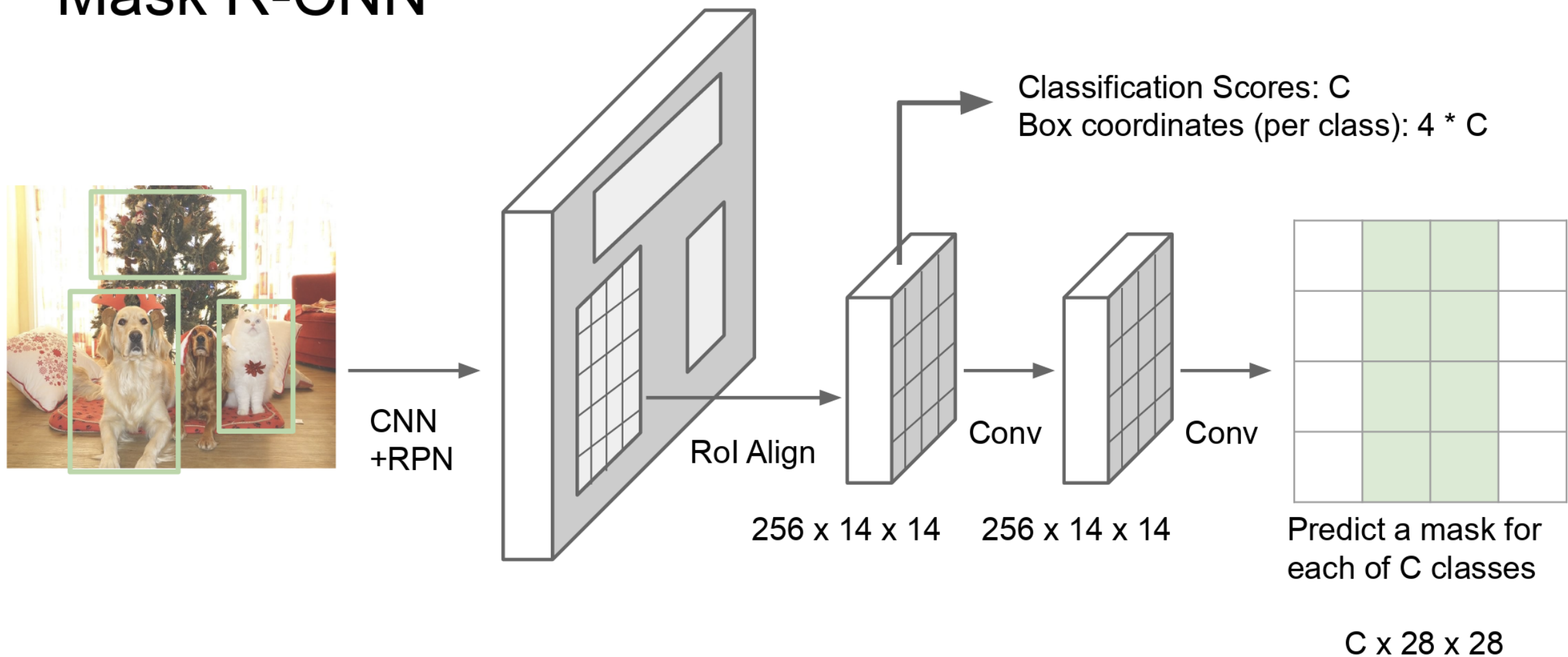
Object Detection: Faster R-CNN



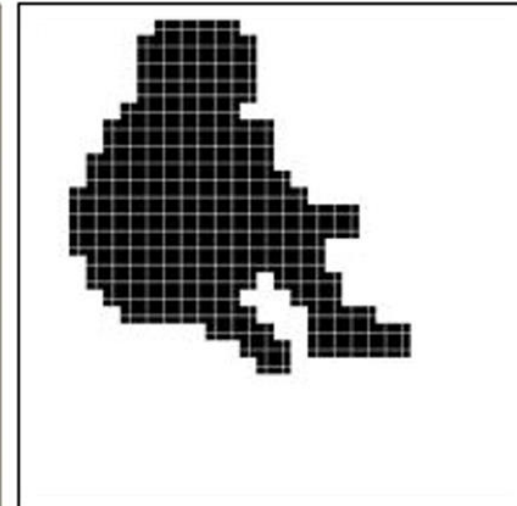
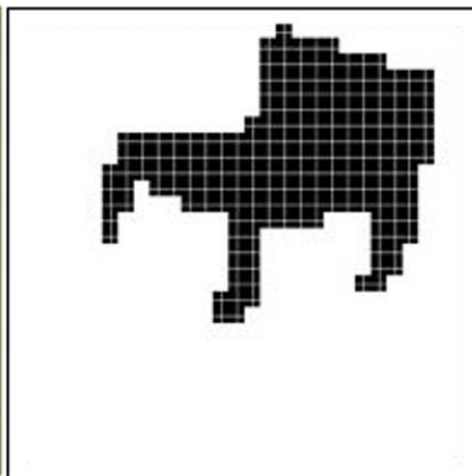
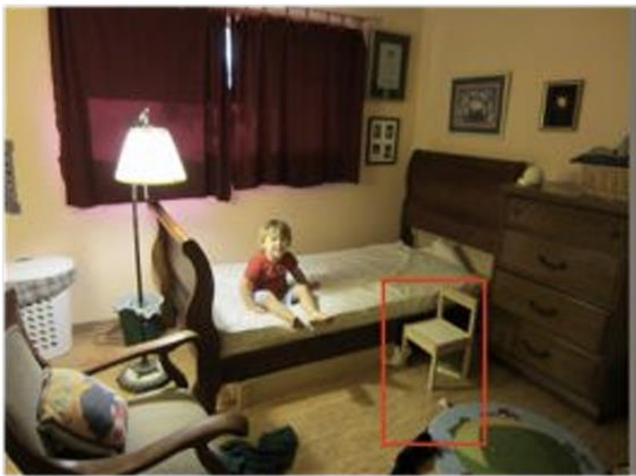
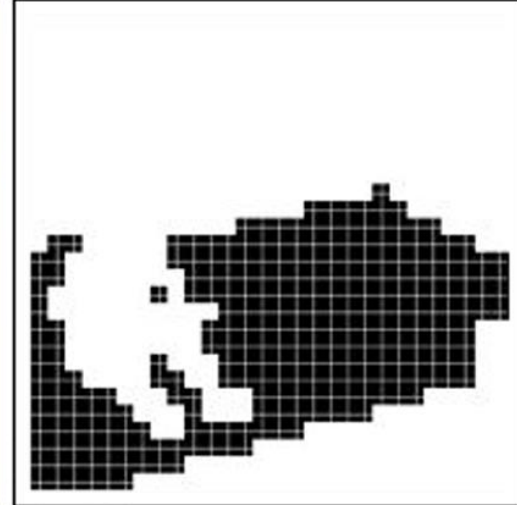
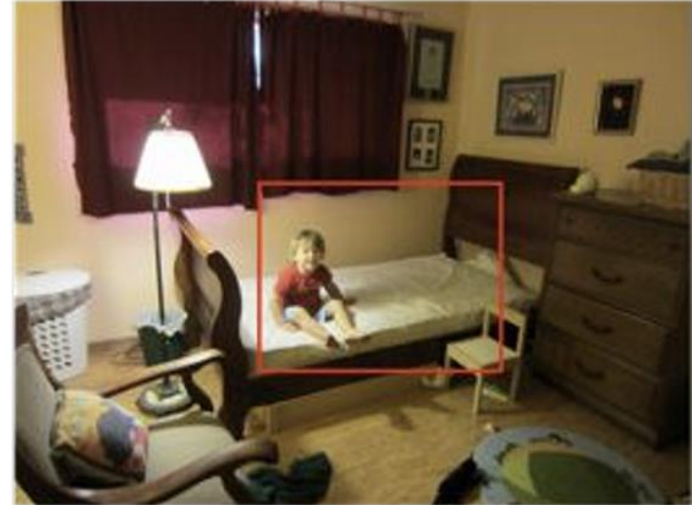
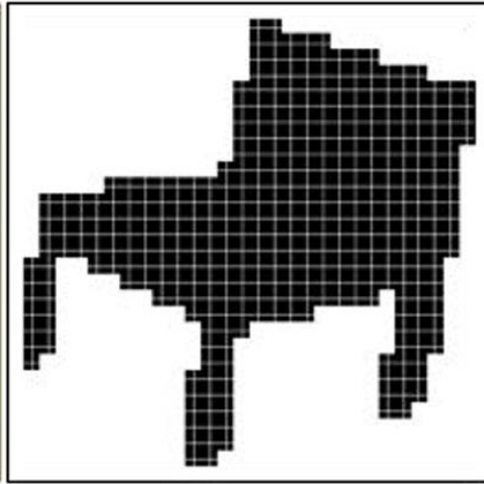
Instance Segmentation: Mask R-CNN



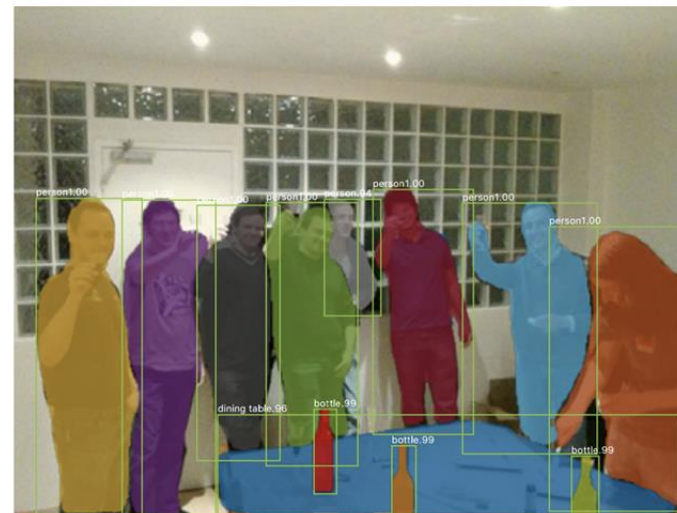
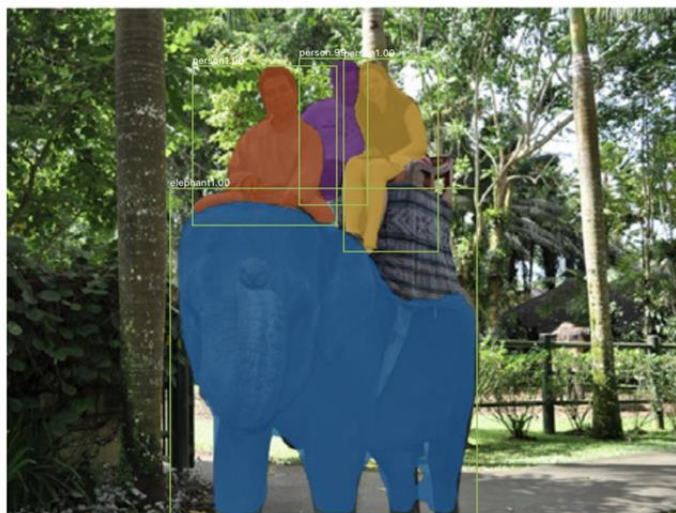
Mask R-CNN



Mask R-CNN: Example Mask Training Targets

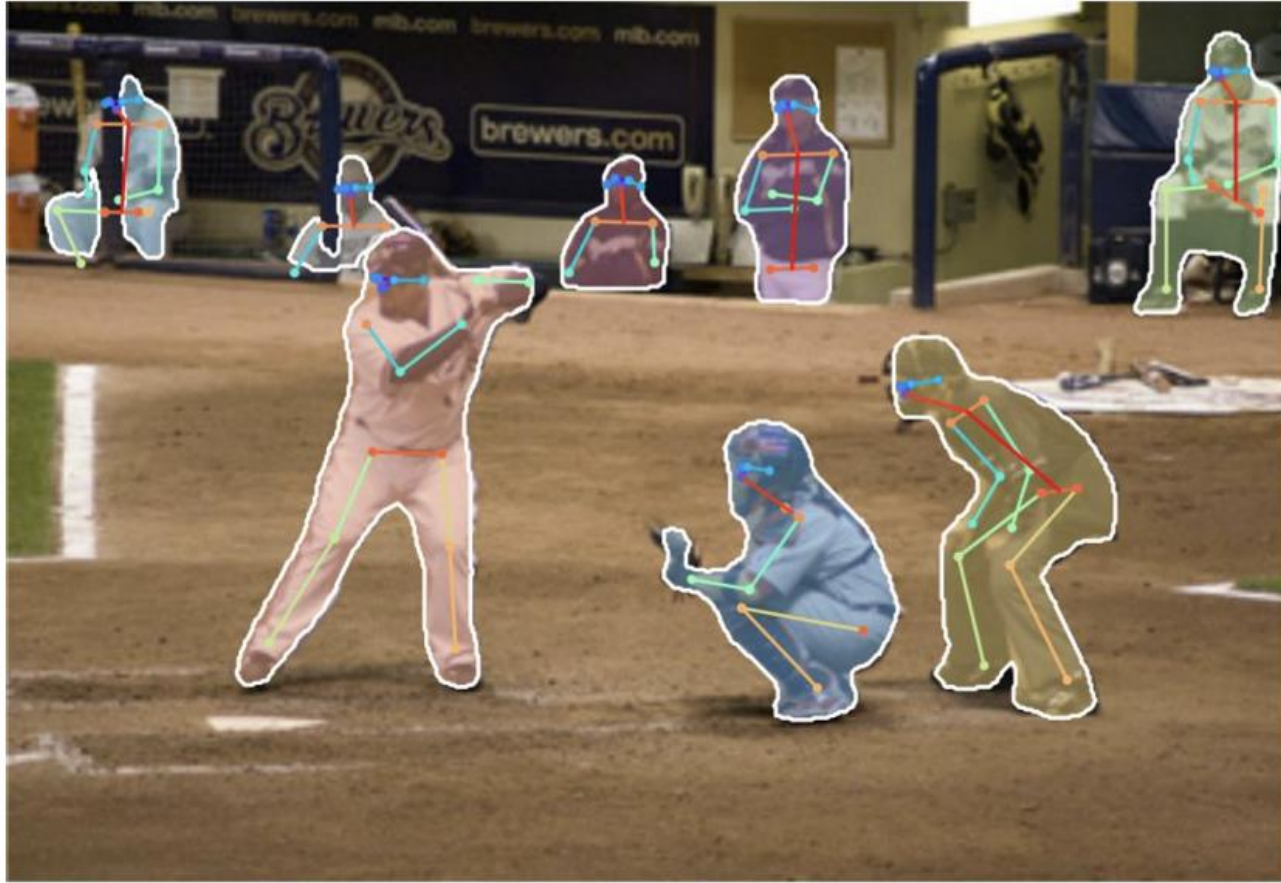


Mask R-CNN: Results



Mask R-CNN

Also does pose



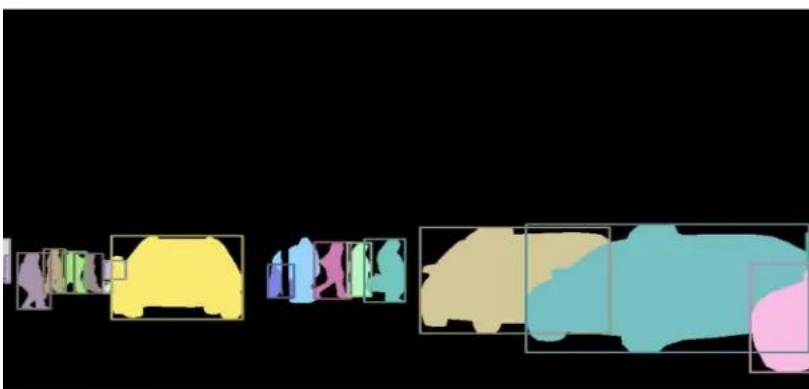
Panoptic Segmentation



(a) image



(b) semantic segmentation



(c) instance segmentation



(d) panoptic segmentation

Panoptic Segmentation

- Combines **semantic + instance segmentation**
- Assigns:
 - **Class label** (what)
 - **Instance ID** (which object)
- Every pixel is uniquely labeled
- Goal: Complete scene understanding

Deep Learning Models for Panoptic Segmentation

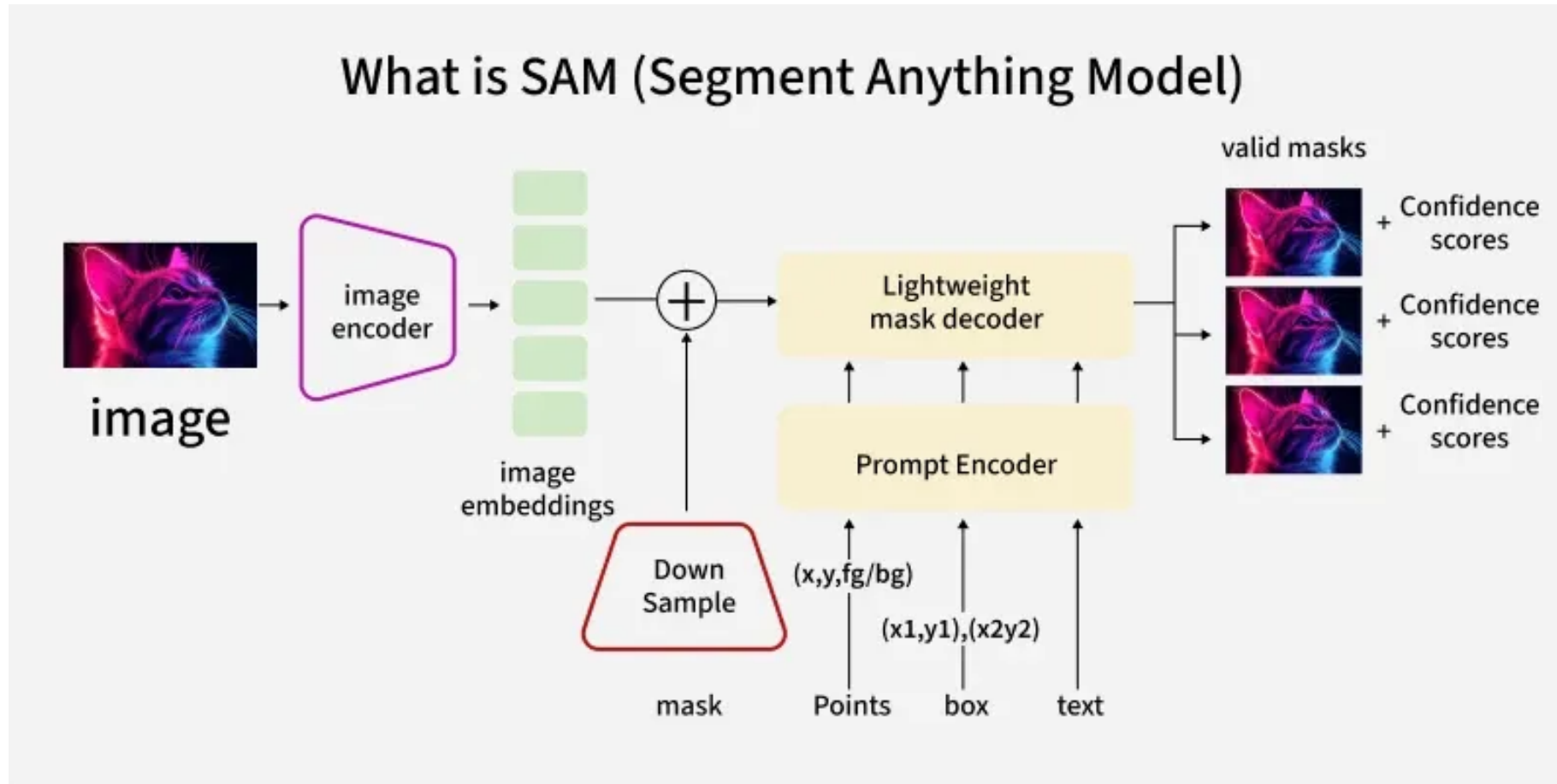
- **Mask2former**: Currently a leading architecture that uses masked attention to achieve state-of-the-art results in panoptic, instance, and semantic segmentation.
- **Panoptic-FPN**: A seminal, efficient model that extends the Feature Pyramid Network (FPN) with separate heads for semantic and instance segmentation to create a joint output.
- **YOLO-based Models (e.g., YOLOv3/YOLO26)**: Modified versions of the YOLO detection model designed to provide high-precision, real-time performance on edge devices.
- **Panoptic Diffusion Models (PDM)**: An emerging class of models that concurrently generate images and their corresponding panoptic segmentation maps, bridging image generation with spatial understanding.

SAM (Segment Anything Model)

- an open source image segmentation model by Meta
- Can segment **any object** in an image without task-specific training
- Works with different types of prompts:
 - Points
 - Bounding boxes
 - Text (via extensions)
- Traditional segmentation models:
 - Require task-specific training
 - Need large labeled datasets
- SAM aims to:
 - Enable zero-shot segmentation
 - Generalize across domains
 - Reduce dependency on manual annotation



SAM (Segment Anything Model)



SAM vs Panoptic Segmentation

Aspect	SAM	Panoptic Segmentation
Task type	Prompt-based	Predefined task
Output	Masks (no labels by default)	Masks + class labels
Instance separation	Yes	Yes
Semantic labeling	 No (by default)	 Yes
Fully automatic	 Needs prompts	 Yes

Evolution of Deep Learning Models for Segmentation Tasks

Model	Year	Key Innovation	Strengths
FCN	2015	Fully conv, upsampling	End-to-end training sciencedirect
U-Net	2015	Skip connections, U-shape	Precise boundaries saiwa
DeepLabv3+	2018	Atrous conv, ASPP	Multi-scale context arxiv
Mask R-CNN	2017	Mask branch on R-CNN	Instance seg arxiv
SAM 2	2024	Promptable ViT encoder	Zero-shot flexibility hiringnet

Reference

- https://cs231n.stanford.edu/slides/2022/lecture_9_jiajun.pdf
- <https://medium.com/data-science/understanding-semantic-segmentation-with-unet-6be4f42d4b47>